

SURGE Project Abstract

Project Title: Efficient neural network for polymer dynamics simulation

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Abstract

In my SURGE project, I am working on improving the efficiency of the simulation of polymer chains using the bead-spring model in Brownian dynamics simulations. In this work, instead of using the normal step by step simulation method, we are trying to use a neural network to predict the motion of polymer beads for larger time steps. For preparing the dataset, equilibrium configurations of a 6-bead and 5-spring FENE chain (Kuhn length, $v = 100$) were generated using sufficiently long simulations. From this, around 2000 equilibrium configurations were selected.

For each equilibrium configuration, 13 more configurations were generated by running the simulation further for various number of timesteps. Note, in this stage, the **Brownian force was turned off** and only spring forces were acting on the beads. These new configurations were stored at different time intervals. For example, first after 5-time steps, then after 10-time steps and so on.

Before training the neural network, preprocessing of data was done. First, the centre of mass of the chain was subtracted from bead positions. After that, normalization was done using half of the maximum chain length, which is 250. In the training data, the initial configuration is taken as input and the configurations after different time steps are used as outputs.

After training, the neural network is expected to directly predict the forward deterministic trajectory for larger time steps. Then a Brownian noise vector is added to include random motion in the system. Owing to this, the simulation can run with much larger time steps and it may help in making simulations faster.